

A208 · Building dynamical models of multi-step state transitions from single cell gene expression trajectories

Presenter: Yukai You — Northeastern University

Authors: Yukai You

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Multi-step cell state transitions often occur in biological processes, such as cell differentiation and disease progression, yet the regulatory mechanisms governing these transitions remain unclear. Here, we introduce NetDes, a computational method that integrates top-down and bottom-up systems biology to infer core transcription factor regulatory networks and build ODE-based dynamical models from single-cell gene expression trajectories. We demonstrate that NetDes predicts regulatory interactions and reproduces gene expression dynamics through benchmarking using in-silico time trajectories with decoys, tests on gene circuit simulations of embryonic phenotypic switching, and application to time-series scRNA-seq data from human iPSC differentiation. Compared to existing approaches, NetDes has the advantage of capturing sequential state transitions within a single dynamical model. Network simulations and coarse-graining further elucidate the regulatory roles of genes and their combinations in driving these transitions. Our approach provides a generalizable framework for mechanistic modeling of gene regulation in complex cell state transitions.